

ARTIFICIAL INTELLIGENCE ASSIGNMENT

Course: Artificial Intelligence (CS-401)

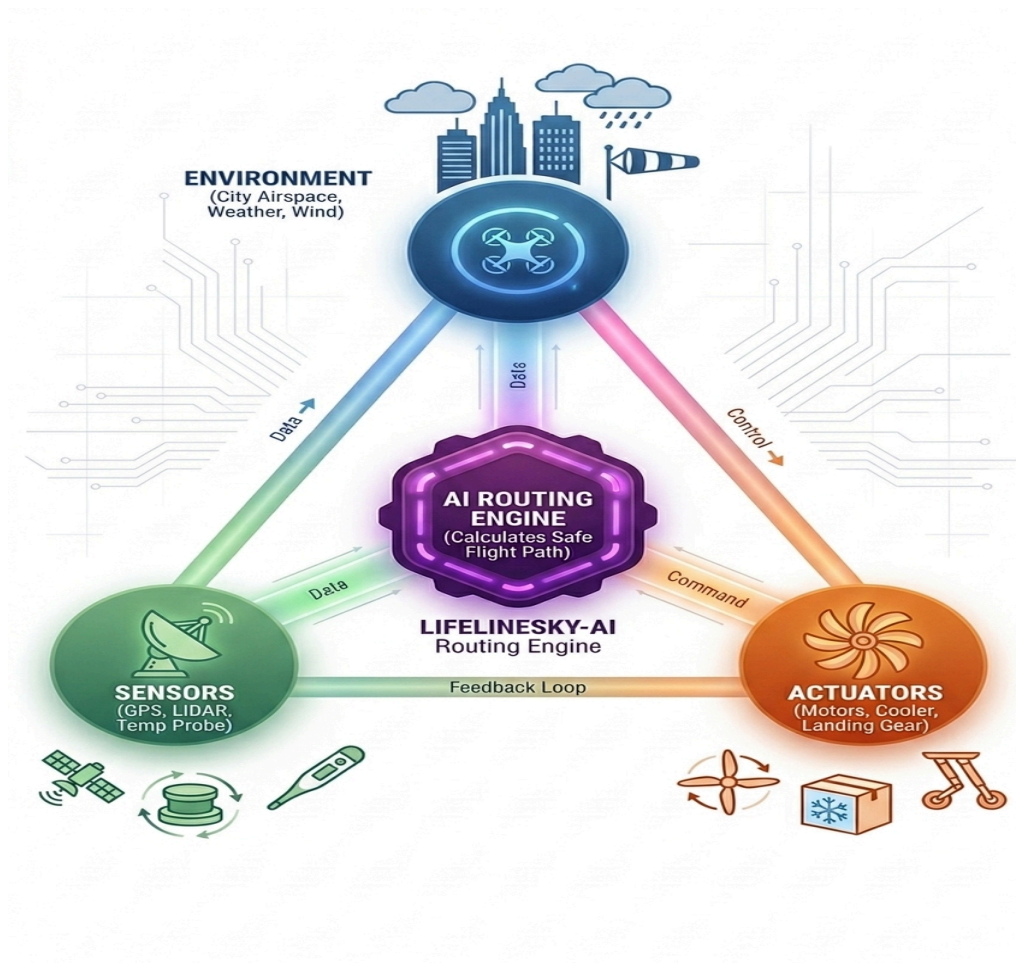
Topic: AI Agent Formulation & PEAS Framework

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1. Agent Name & Overview

- **Agent Name:** LifelineSky-AI
- **Type of Agent:** Autonomous Medical Drone Routing Agent
- **Simple Overview:** LifelineSky-AI is a smart drone system designed to deliver emergency medical supplies like blood bags, organs, and vaccines between hospitals and emergency centers. It avoids ground traffic gridlock to deliver life-saving supplies quickly while keeping the medicine cold and safe.

2. Graphical Architecture Diagram



3. Agent Formulation

A. Sensors (Inputs)

- **GPS & Altitude Sensor:** Tracks the current position and height of the drone.
- **LIDAR & Camera:** Scans the surroundings for buildings, power lines, and birds to prevent accidents.
- **Medicine Temperature Probe:** Monitors the box temperature to make sure supplies do not spoil.
- **Wind & Weather Sensor:** Measures wind speed and rain so the drone can adjust its flight.

B. Actuators (Outputs)

- **Rotor Motors:** Controls speed, direction, and turning during flight.
- **Cooling Box System:** Keeps the medicine inside at a safe, low temperature.
- **Landing Gear:** Drops down to allow safe landing on hospital helipads.
- **Warning Lights & Beeper:** Alerts people on the ground during takeoff and landing.

C. Environment Type

- **Partially Observable:** The drone cannot see behind large buildings or predict sudden weather changes until its sensors detect them.
- **Dynamic:** Weather, wind, and obstacles in the air are constantly changing.
- **Continuous:** Flight speed, altitude, and temperature are continuous values.
- **Stochastic (Unpredictable):** Random events like sudden wind gusts or flying birds can happen anytime.

4. PEAS Framework Evaluation

PEAS Element	Description
Performance Measure	1. Deliver medical packages quickly and safely. 2. Keep blood and organs at the correct cool temperature. 3. Avoid collisions with buildings or birds. 4. Save battery power for long flights.
Environment	City airspace, hospital rooftops, unpredictable wind/rain, tall buildings, and landing pads.

Actuators	Flight motors, cooling box system, landing gear, warning lights, status display.
Sensors	GPS receiver, obstacle LIDAR/camera, weather sensors, temperature probe, battery level indicator.

5. Conclusion

LifelineSky-AI shows how a simple rational agent can solve a big real-world problem. By reading sensor inputs like location, weather, and temperature, the agent makes smart flight decisions to deliver life-saving medical supplies faster than ground ambulances.